Topic: Review of Differential Equations

I. Introduction

Lets begin by considering the Mth order differential equation of the form:

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1 y' + a_0 y = f(t) \tag{1}$$

where $M \in \mathbf{N}$; the coefficients $a_0, a_1, a_2, ..., a_{M-2}, a_{M-1} \in \mathbf{R}$ and has the solution y (function of t).

Ex: Examples of this type of an equation would be:

$$y^{(5)} + 3y'' = \sin t \tag{2}$$

where $M =$ $\quad$ $, a_4 =$ $\quad$ $, a_3 =$ $\quad$ $, a_2 =$ $\quad$ $, a_1 =$ $\quad$ $a_0 =$ $\quad$, and $f(t) =$

$$3y''' + 6y'' - 14y = 0 \quad \Longrightarrow \tag{3}$$

where $M =$ $\quad$ $, a_2 =$ $\quad$ $, a_1 =$ $\quad$ $a_0 =$ $\quad$, and $f(t) =$

Defn: The differential equation in (1) is said to be **homogeneous** if . It is called **non-homogeneous** otherwise.

Ex: Equation (2) is

Equation (3) is

<u>Ex</u>: Logically solve the following equations by filling in the blanks

(1) $4y = 3t^2$: $\quad 4\Big(\qquad \Big) = 3t^2 \quad \Longrightarrow \qquad\qquad$ is a solution of (1)

(2) $y' = 3t^2$: $\quad \Big(\qquad \Big)' = 3t^2 \quad \Longrightarrow \qquad\qquad$ is a solution of (2)

(3) $y' + 4y = 0$: $\quad \Big(\qquad \Big)' + 4\Big(\qquad \Big) = 0$

$\Longrightarrow \qquad\qquad$ is a solution of (3)

(4) $y' + 4y = 3t^2$: $\quad \Big(\qquad\qquad \Big)' + 4\Big(\qquad\qquad \Big) = 3t^2$

$\Longrightarrow \qquad\qquad$ is a solution of (4)

Important Question: Did we find all the solutions to the equations in the last example?

(1) $4y = 3t^2$: Is $y = \frac{3}{4}t^2$ the only solution?

(2) $y' = 3t^2$: Is $y = t^3 + c$ the only solution?

(3) $y' + 4y = 0$: Is $y = e^{-4t} + c$ the only solution?

(4) $y' + 4y = 3t^2$: Is $y = \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c$ the only solution?

Let $y_3 = e^{-4t} + c$ be the solution to Eqn 3 (i.e.) and

$y_4 = \frac{3}{4}t^2 - \frac{3}{8}t + \frac{3}{32} + c$ be the solution to Eqn 4 (i.e.)

and consider the new function

$$y =$$

Plug y into Eqn (4):

$$y' + 4y$$

$\Longrightarrow$ is also a solution of (4)

Follow-up Question: What does this tell us about solutions to the differential equation below?

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1 y' + a_0 y = f(t)$$

It says that the complete solution, $y(t)$, comes from the sum of solutions of the 2 equations below:

(1) $y^{(M)} + a_{M-1}y^{(M-1)} + \cdots + a_1 y' + a_0 y =$ (*homogeneous*)

(2) $y^{(M)} + a_{M-1}y^{(M-1)} + \cdots + a_1 y' + a_0 y =$ (*non − homogeneous*)

Defn: The solution of eqn (1) is called the and

The solution of eqn (2) is called the

Thm: The general solution, y, of

$$y^{(M)} + a_{M-1}y^{(M-1)} + a_{M-2}y^{(M-2)} + \cdots + a_1 y' + a_0 y = f(t)$$

is given by $y =$

Notes: (1) The strategy we used to "guess" the form of the particular solution is known as the method of

(2) The form of y_p we "guess" should not contain a constant multiple of any

Ex: Find the solution of $y'' + 3y' - 4y = 4t + 5$; with $y(0) = 0$; $y'(0) = -1$; and given that

$y_h(t) = c_1 e^t + c_2 e^{-4t}$

Want:

Strategy:
(1) Guess
(2) Compute
(3) Plug
(4) Group
(5) Solve
(6) Sub
(7) Apply

(1) Guess:

(2) Compute:

(3) Plug:

(4) Group:

(5) Solve:

(6) Sub:

(7) Solve:

What would the guess for y_p have been if we were instead solving $y'' + 3y' = 4t + 5$?

Ex: Find the particular solution of $y'' - 4y = 2e^{3t}$ given that we know $y_h = c_1e^{2t} + c_2e^{-2t}$

Guess:

Compute:

Plug:

Group:

Solve:

Ex: Find the particular solution of $y'' - 4y = 2e^{2t}$ given that we know $y_h = c_1e^{2t} + c_2e^{-2t}$

Guess:

Compute:

Plug:

Group:

Solve:

Ex: Find the solution of $3y'' + y' - 2y = 2\cos t;\ \ y(0) = \frac{8}{13};\ y'(0) = -\frac{14}{13};$ and $y_h = c_1 e^{\frac{2}{3}t} + c_2 e^{-t}$

Guess:

II. The Homogeneous Solution, y_h, for Mth Order Linear DEs

Consider the Mth order homogeneous differential equation

(1) $$a_M y^{(M)} + a_{M-1} y^{(M-1)} + a_{M-2} y^{(M-2)} + \cdots + a_1 y' + a_0 y = 0$$

Defn: The Mth degree polynomial

$$a_M r^M + a_{M-1} r^{M-1} + a_{M-2} r^{M-2} + \cdots + a_1 r + a_0 = 0$$

is called the **characteristic equation** of (1)

Thm: If $r_1, r_2, ..., r_M$ are the roots of the characteristic equation then the solution of (1) is

$$y(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t} + \cdots + A_{M-1} e^{r_{M-1} t} + A_M e^{r_M t}$$

where $A_1, A_2, ...A_{M-1}, A_M$ are constants.

Note: The roots of the characteristic equation can be complex.

III. The Homogeneous Solution, y_h, for 2nd Order Linear DEs

Consider the 2nd order homogeneous differential equation

(1) $$a_2 y'' + a_1 y' + a_0 y = 0$$

which has the the characteristic equation

with roots: r_1, r_2

The possible forms of the solution to (1) can be summarized in the following table:

If	Then

Ex: Find the solution of $y'' + 4y = 3t^2$

Ex: Find the general solution of $3y'' + y' - 2y = 2\cos t$